

Problem

Part 1

In Part 1, the problem was to take the accelerometer data and display it in seconds instead of the recorded milliseconds. The output also needed to show the x, y, and z acceleration values, with certain formatting and decimal units of accuracy, the seconds had a similar constraint.

Part 2

In Part 2, the problem given to us was to write a function called `mag()` that would calculate the magnitude of the acceleration. This meant directing sensor inputs from the controller into a formulated equation used to find magnitude and `<math.h>` commands to write that formula.

Part 3

In Part 3, the task was to take the time given in milliseconds and format it into easier to read units. So it was asked to come up with functions to convert the time given into minutes, seconds, and milliseconds, with remainders for each one.

Part 4

In Part 4, the problem changed focus from the accelerometer to the controller's buttons, specifically the ones with shapes. The goal was to write a new program that could count the amount of the four shaped buttons were being pressed at once. Solving the problem required creating a new function that returned this count without using `scanf()` or `printf()` inside of it as well as altering the `while()` function.

Analysis

Part 1

The input for Part 1 was milliseconds and the three acceleration values from the controller's sensors. The output was "Echoing output: seconds and the accelerations", all printed with precise formatting. This part seemed to be geared towards testing the ability to handle numerical formatting in C.

Part 2

The input for Part 2 was the same accelerometer values, but the output focused now on the acceleration values rather than the time, by finding the acceleration's magnitude. The formula used was the square root of x squared plus y squared plus z squared. showing

how mathematical functions can be used to process sensor data, as well as testing knowledge of `<math.h>`.

Part 3

For Part 3, the input was just the time in milliseconds, but we needed the output to be in minutes, seconds, and milliseconds, along with the previous magnitude. This required integer division and modulo operators to convert the time.

Part 4

In Part 4, the input was four integers representing pressing each of the four shaped buttons. The output was one integer that represented the total number of buttons pressed at once. This part focused on problem solving, by using outside knowledge to solve a problem in a unique fashion.

Design

Part 1

In Part 1, I used the provided code, while modifying the print statement to include specifiers that made sure to print the data in the specific character and digit parameters. I also divided the "t" variable by 1000.0 to convert them into seconds, creating a new double variable called seconds.

Part 2

In Part 2, I wrote the `mag()` function. It took three doubles as inputs and returned the square root of their squares added together. I included `math.h` at the top and used `-lm` when compiling so the math library would be linked correctly.

Part 3

In Part 3, I added three new functions: `minutes()`, `seconds()`, and `millis()`. Each one used integer division or modulus to return the correct part of the time. This made the output much easier to read and understand.

Part 4

In Part 4, I copied my previous program and but commented the old code and cleared the while variable. I made new variables for each one of the buttons, then I created a new function called "`count_pressed()`", that added up added up the four inputs, when they weren't zero. The main function read the inputs, sent them to the new function I made, and then printed the result.

Testing

Part 1

For Part 1, I tested by running the program and checking that the seconds displayed with three decimal places and that the accelerometer values matched the format requirements.

Part 2

For Part 2, I tested by moving the controller around and watching the magnitude. When the controller was still, the magnitude was close to one, and when I tilted it, the value changed as expected.

Part 3

For Part 3, I tested by letting the program run long enough to produce large millisecond values. I confirmed that values like 129313 ms were properly displayed as 2 minutes, 9 seconds, and 313 milliseconds.

Part 4

For Part 4, I tested by pressing different combinations of the shape buttons. The output showed 0 when no buttons were pressed, 1 or 2 when one or two buttons were pressed, and 4 when all four buttons were pressed at the same time.

Comments

Through all four parts of the lab, I learned practical ways to use functions, as well as truly learning how much more organized they make code. I also became a little more comfortable with using format specifiers and the math library. Part 4 was very interesting, I was extremely confused at first on how to solve it, which is when I turned to google for some help which taught me what the “!” (not) operator was. But when I talked with the TA’s I found that I used a different method from most, most used “if/else”, I wish I used this because I feel it would’ve been a lot easier and more useful.

Implementation

The following shows my completed code for lab3.c (Parts 1–3):

```
#include <stdio.h>

#include <math.h>

double mag(double x, double y, double z) {

    return sqrt(x*x + y*y + z*z);

}

int minutes(int t) {

    return t / 60000;

}

int seconds(int t) {

    return (t / 1000) % 60;

}

int millis(int t) {

    return t % 1000;

}

int main(void) {

    int t;

    double ax, ay, az;

    while (1) {

        scanf("%d,%lf,%lf,%lf", &t, &ax, &ay, &az);

/* CODE SECTION 0 */

        double seconds = t / 1000.0;

        printf("Echoing output: %8.3f, %7.4f, %7.4f, %7.4f\n", seconds, ax, ay, az); */

/* CODE SECTION 1 */
```

```

        printf("At %d ms, the acceleration's magnitude was: %lf\n",
               t, mag(ax, ay, az));
/* CODE SECTION 2 */

        printf("At %d minutes, %d seconds, and %d milliseconds it was: %lf\n",
               minutes(t), seconds(t), millis(t), mag(ax,ay,az));
    }
    return 0;
}

```

The following shows my completed code for lab3-2.c (Part 4):

```

#include <stdio.h>

#include <math.h>

int count_pressed(int triangle, int circle, int cross, int square) {
    return (triangle != 0) + (circle != 0) + (cross != 0) + (square != 0);
}

int main(void) {
    int tri, cir, cro, squ;

    while (1) {
        if (scanf("%d,%d,%d,%d", &tri, &cir, &cro, &squ) != 4) {
            continue;
        }

        printf("%d\n", count_pressed(tri, cir, cro, squ));

        fflush(stdout);
    }

    return 0;
}

```

Output

Part 1:

```
simons@WKCD2048-18 /cygdrive/d/CPRE1850/Lab3
$ ./ds4rd.exe -d 054c05c4 -D D54_BT -t e -l ./lab3.e
Echoing output: 0.030, 0.0436, 0.9874, 0.1962, 0.1962
Echoing output: 0.046, 0.0432, 0.9823, 0.1960
Echoing output: 0.065, 0.0443, 0.9847, 0.1971
Echoing output: 0.080, 0.0455, 0.9867, 0.1975
Echoing output: 0.097, 0.0467, 0.9854, 0.1962
Echoing output: 0.113, 0.0441, 0.9854, 0.1938
Echoing output: 0.129, 0.0427, 0.9854, 0.1971
Echoing output: 0.146, 0.0467, 0.9871, 0.1970
Echoing output: 0.164, 0.0473, 0.9867, 0.1984
Echoing output: 0.181, 0.0466, 0.9861, 0.1982
Echoing output: 0.196, 0.0391, 0.9905, 0.1993
Echoing output: 0.215, 0.0410, 0.9849, 0.1958
Echoing output: 0.230, 0.0467, 0.9849, 0.1981
Echoing output: 0.246, 0.0446, 0.9869, 0.1958
Echoing output: 0.263, 0.0457, 0.9835, 0.1994
Echoing output: 0.291, 0.0454, 0.9818, 0.1928
Echoing output: 0.303, 0.0475, 0.9900, 0.1938
Echoing output: 0.314, 0.0482, 0.9913, 0.1955
Echoing output: 0.330, 0.0430, 0.9879, 0.1956
Echoing output: 0.346, 0.0432, 0.9854, 0.1953
Echoing output: 0.373, 0.0450, 0.9844, 0.1917
Echoing output: 0.396, 0.0419, 0.9827, 0.1966
Echoing output: 0.413, 0.0446, 0.9905, 0.1977
Echoing output: 0.430, 0.0465, 0.9805, 0.1954
Echoing output: 0.446, 0.0446, 0.9854, 0.1968
Echoing output: 0.462, 0.0408, 0.9849, 0.1958
Echoing output: 0.480, 0.0451, 0.9880, 0.1978
Echoing output: 0.495, 0.0419, 0.9874, 0.1977
Echoing output: 0.513, 0.0421, 0.9817, 0.1973
Echoing output: 0.530, 0.0435, 0.9841, 0.1989
Echoing output: 0.547, 0.0414, 0.9862, 0.1956
Echoing output: 0.563, 0.0449, 0.9874, 0.1979
Echoing output: 0.581, 0.0444, 0.9844, 0.1968
Echoing output: 0.596, 0.0461, 0.9897, 0.1956
Echoing output: 0.613, 0.0425, 0.9852, 0.1975
Echoing output: 0.629, 0.0468, 0.9819, 0.1982
Echoing output: 0.656, 0.0465, 0.9857, 0.1966
Echoing output: 0.680, 0.0456, 0.9883, 0.1955
Echoing output: 0.696, 0.0440, 0.9838, 0.1961
Echoing output: 0.713, 0.0447, 0.9838, 0.1954
Echoing output: 0.730, 0.0447, 0.9863, 0.1984
Echoing output: 0.748, 0.0438, 0.9866, 0.1960
Echoing output: 0.763, 0.0433, 0.9806, 0.1960
Echoing output: 0.780, 0.0443, 0.9844, 0.1983
Echoing output: 0.797, 0.0440, 0.9851, 0.1989
Echoing output: 0.814, 0.0428, 0.9880, 0.1978
Echoing output: 0.831, 0.0421, 0.9836, 0.1976
Echoing output: 0.847, 0.0469, 0.9845, 0.2006
Echoing output: 0.864, 0.0430, 0.9910, 0.1953
Echoing output: 0.879, 0.0434, 0.9861, 0.2005
Echoing output: 0.897, 0.0440, 0.9858, 0.1968
Echoing output: 0.914, 0.0436, 0.9911, 0.1995
Echoing output: 0.929, 0.0436, 0.9886, 0.2000
Echoing output: 0.947, 0.0424, 0.9877, 0.1968
Echoing output: 0.963, 0.0417, 0.9883, 0.1978
Echoing output: 0.980, 0.0416, 0.9907, 0.1964
Echoing output: 0.999, 0.0402, 0.9893, 0.1988
Echoing output: 1.014, 0.0407, 0.9900, 0.1942
Echoing output: 1.030, 0.0429, 0.9871, 0.1961
Echoing output: 1.047, 0.0423, 0.9863, 0.1968
Echoing output: 1.064, 0.0468, 0.9900, 0.1977
Echoing output: 1.081, 0.0392, 0.9860, 0.1947
Echoing output: 1.096, 0.0428, 0.9880, 0.1971
Echoing output: 1.113, 0.0418, 0.9883, 0.1938
Echoing output: 1.130, 0.0421, 0.9893, 0.1948
Echoing output: 1.147, 0.0423, 0.9868, 0.1981
Echoing output: 1.163, 0.0467, 0.9855, 0.1959
Echoing output: 1.181, 0.0446, 0.9816, 0.1965
Echoing output: 1.197, 0.0418, 0.9845, 0.1955
Echoing output: 1.212, 0.0427, 0.9861, 0.1967
Echoing output: 1.231, 0.0435, 0.9862, 0.2026
Echoing output: 1.249, 0.0425, 0.9860, 0.1987
Echoing output: 1.263, 0.0430, 0.9871, 0.1949
Echoing output: 1.281, 0.0441, 0.9875, 0.2001
Echoing output: 1.297, 0.0428, 0.9883, 0.1940
Echoing output: 1.314, 0.0465, 0.9844, 0.1983
Echoing output: 1.330, 0.0446, 0.9849, 0.1993
Echoing output: 1.347, 0.0443, 0.9893, 0.1955
Echoing output: 1.364, 0.0428, 0.9862, 0.1982
Echoing output: 1.380, 0.0441, 0.9846, 0.1983
Echoing output: 1.401, 0.0436, 0.9902, 0.1997
Echoing output: 1.419, 0.0435, 0.9875, 0.1971
Echoing output: 1.438, 0.0428, 0.9880, 0.1958
Echoing output: 1.450, 0.0458, 0.9911, 0.1962
Echoing output: 1.465, 0.0423, 0.9839, 0.1951
Echoing output: 1.483, 0.0414, 0.9862, 0.1954
Echoing output: 1.498, 0.0474, 0.9845, 0.1956
Echoing output: 1.516, 0.0416, 0.9856, 0.1977
Echoing output: 1.531, 0.0443, 0.9846, 0.1968
Echoing output: 1.553, 0.0471, 0.9907, 0.1958
Echoing output: 1.567, 0.0510, 0.9855, 0.1954
Echoing output: 1.582, 0.0425, 0.9883, 0.2001
Echoing output: 1.602, 0.0422, 0.9877, 0.1954
Echoing output: 1.617, 0.0434, 0.9891, 0.1975
Echoing output: 1.636, 0.0484, 0.9856, 0.1979
Echoing output: 1.651, 0.0458, 0.9794, 0.1962
Echoing output: 1.666, 0.0418, 0.9856, 0.1958
Echoing output: 1.688, 0.0440, 0.9886, 0.1975
Echoing output: 1.700, 0.0455, 0.9915, 0.2001
Echoing output: 1.720, 0.0452, 0.9874, 0.1971
Echoing output: 1.734, 0.0501, 0.9823, 0.1955
Echoing output: 1.749, 0.0452, 0.9905, 0.1953
Echoing output: 1.764, 0.0467, 0.9821, 0.1977
Echoing output: 1.781, 0.0463, 0.9847, 0.1978
Echoing output: 1.797, 0.0429, 0.9818, 0.1960
Echoing output: 1.815, 0.0423, 0.9886, 0.1933
Echoing output: 1.833, 0.0427, 0.9880, 0.1947
Echoing output: 1.849, 0.0447, 0.9877, 0.1968
Echoing output: 1.865, 0.0429, 0.9868, 0.1954
Echoing output: 1.881, 0.0436, 0.9827, 0.1967
Echoing output: 1.898, 0.0422, 0.9880, 0.1976
Echoing output: 1.914, 0.0411, 0.9866, 0.1947
Echoing output: 1.931, 0.0482, 0.9877, 0.1988
Echoing output: 1.948, 0.0477, 0.9868, 0.1998
Echoing output: 1.964, 0.0421, 0.9871, 0.1989
Echoing output: 1.981, 0.0407, 0.9843, 0.1940
Echoing output: 1.997, 0.0443, 0.9864, 0.1984
Echoing output: 2.014, 0.0452, 0.9841, 0.1953
Echoing output: 2.032, 0.0427, 0.9886, 0.1929
Echoing output: 2.047, 0.0455, 0.9883, 0.1979
Echoing output: 2.065, 0.0400, 0.9812, 0.1943
Echoing output: 2.080, 0.0427, 0.9839, 0.1954
Echoing output: 2.098, 0.0422, 0.9884, 0.1977
Echoing output: 2.115, 0.0407, 0.9858, 0.1979
Echoing output: 2.130, 0.0460, 0.9874, 0.1954
Echoing output: 2.150, 0.0446, 0.9894, 0.1984
Echoing output: 2.166, 0.0408, 0.9871, 0.1977
```

Part 2:

```
simon@1920248-18 /cydrive/d/CPRE1850/Lab3
$ ./ds4rd.exe -d 054c05c4 -D DS4_BT -t -g | ./lab3.exe
At 16 ms, the acceleration's magnitude was: 0.998646
At 32 ms, the acceleration's magnitude was: 1.006004
At 48 ms, the acceleration's magnitude was: 1.014831
At 66 ms, the acceleration's magnitude was: 1.022241
At 84 ms, the acceleration's magnitude was: 1.005526
At 99 ms, the acceleration's magnitude was: 0.997247
At 116 ms, the acceleration's magnitude was: 1.007943
At 134 ms, the acceleration's magnitude was: 1.016536
At 149 ms, the acceleration's magnitude was: 1.014561
At 167 ms, the acceleration's magnitude was: 1.005200
At 182 ms, the acceleration's magnitude was: 0.998093
At 201 ms, the acceleration's magnitude was: 0.998577
At 218 ms, the acceleration's magnitude was: 1.010042
At 232 ms, the acceleration's magnitude was: 1.013890
At 250 ms, the acceleration's magnitude was: 1.001817
At 266 ms, the acceleration's magnitude was: 0.997692
At 281 ms, the acceleration's magnitude was: 1.006873
At 299 ms, the acceleration's magnitude was: 1.007664
At 315 ms, the acceleration's magnitude was: 1.013632
At 338 ms, the acceleration's magnitude was: 1.011221
At 351 ms, the acceleration's magnitude was: 1.003093
At 366 ms, the acceleration's magnitude was: 1.007077
At 381 ms, the acceleration's magnitude was: 1.005711
At 398 ms, the acceleration's magnitude was: 1.006030
At 417 ms, the acceleration's magnitude was: 1.010144
At 431 ms, the acceleration's magnitude was: 1.004382
At 450 ms, the acceleration's magnitude was: 1.007134
At 465 ms, the acceleration's magnitude was: 1.007159
At 483 ms, the acceleration's magnitude was: 1.008346
At 500 ms, the acceleration's magnitude was: 1.006748
At 515 ms, the acceleration's magnitude was: 1.007860
At 532 ms, the acceleration's magnitude was: 1.013033
At 548 ms, the acceleration's magnitude was: 1.009813
At 568 ms, the acceleration's magnitude was: 1.007674
At 589 ms, the acceleration's magnitude was: 1.008159
At 601 ms, the acceleration's magnitude was: 1.007426
At 616 ms, the acceleration's magnitude was: 1.006606
At 631 ms, the acceleration's magnitude was: 1.009130
At 649 ms, the acceleration's magnitude was: 1.009176
At 666 ms, the acceleration's magnitude was: 1.006717
At 681 ms, the acceleration's magnitude was: 1.002297
At 699 ms, the acceleration's magnitude was: 1.007014
At 716 ms, the acceleration's magnitude was: 1.003952
At 734 ms, the acceleration's magnitude was: 1.011843
At 749 ms, the acceleration's magnitude was: 1.002474
At 765 ms, the acceleration's magnitude was: 1.006669
At 783 ms, the acceleration's magnitude was: 1.009791
At 799 ms, the acceleration's magnitude was: 1.009433
At 816 ms, the acceleration's magnitude was: 1.009300
At 833 ms, the acceleration's magnitude was: 1.007005
At 849 ms, the acceleration's magnitude was: 1.004044
At 867 ms, the acceleration's magnitude was: 1.002909
At 883 ms, the acceleration's magnitude was: 1.004978
At 898 ms, the acceleration's magnitude was: 1.006845
At 917 ms, the acceleration's magnitude was: 1.013099
At 934 ms, the acceleration's magnitude was: 1.009038
At 950 ms, the acceleration's magnitude was: 1.007255
At 965 ms, the acceleration's magnitude was: 1.006948
At 984 ms, the acceleration's magnitude was: 1.004010
At 999 ms, the acceleration's magnitude was: 1.004326
At 1015 ms, the acceleration's magnitude was: 1.007288
At 1032 ms, the acceleration's magnitude was: 1.002804
At 1048 ms, the acceleration's magnitude was: 1.005222
At 1063 ms, the acceleration's magnitude was: 1.009427
At 1083 ms, the acceleration's magnitude was: 1.005440
At 1098 ms, the acceleration's magnitude was: 1.006208
At 1115 ms, the acceleration's magnitude was: 1.012391
At 1132 ms, the acceleration's magnitude was: 1.003091
At 1148 ms, the acceleration's magnitude was: 1.004635
At 1167 ms, the acceleration's magnitude was: 1.003794
At 1183 ms, the acceleration's magnitude was: 1.006892
At 1201 ms, the acceleration's magnitude was: 1.009707
At 1215 ms, the acceleration's magnitude was: 1.006889
At 1232 ms, the acceleration's magnitude was: 1.006172
At 1249 ms, the acceleration's magnitude was: 1.006619
At 1266 ms, the acceleration's magnitude was: 1.001293
At 1282 ms, the acceleration's magnitude was: 1.003140
At 1299 ms, the acceleration's magnitude was: 1.007112
At 1315 ms, the acceleration's magnitude was: 1.006516
At 1332 ms, the acceleration's magnitude was: 1.009907
At 1355 ms, the acceleration's magnitude was: 1.006261
At 1368 ms, the acceleration's magnitude was: 1.003725
At 1383 ms, the acceleration's magnitude was: 1.005300
At 1403 ms, the acceleration's magnitude was: 1.008091
At 1417 ms, the acceleration's magnitude was: 1.010482
At 1433 ms, the acceleration's magnitude was: 1.004200
At 1448 ms, the acceleration's magnitude was: 1.005264
At 1471 ms, the acceleration's magnitude was: 1.006375
At 1485 ms, the acceleration's magnitude was: 1.005298
At 1504 ms, the acceleration's magnitude was: 1.004555
At 1518 ms, the acceleration's magnitude was: 1.005132
At 1537 ms, the acceleration's magnitude was: 1.008063
At 1556 ms, the acceleration's magnitude was: 1.002881
At 1568 ms, the acceleration's magnitude was: 1.006832
At 1587 ms, the acceleration's magnitude was: 1.008263
At 1606 ms, the acceleration's magnitude was: 1.006520
At 1619 ms, the acceleration's magnitude was: 1.004622
At 1634 ms, the acceleration's magnitude was: 1.007995
At 1649 ms, the acceleration's magnitude was: 1.008108
At 1668 ms, the acceleration's magnitude was: 1.005158
At 1684 ms, the acceleration's magnitude was: 1.008101
At 1704 ms, the acceleration's magnitude was: 1.008039
At 1721 ms, the acceleration's magnitude was: 1.008216
At 1734 ms, the acceleration's magnitude was: 1.004654
At 1749 ms, the acceleration's magnitude was: 1.006885
At 1767 ms, the acceleration's magnitude was: 1.000752
At 1787 ms, the acceleration's magnitude was: 1.007915
At 1806 ms, the acceleration's magnitude was: 1.008341
At 1817 ms, the acceleration's magnitude was: 1.005770
At 1834 ms, the acceleration's magnitude was: 1.008750
At 1850 ms, the acceleration's magnitude was: 1.007903
At 1867 ms, the acceleration's magnitude was: 1.008320
At 1882 ms, the acceleration's magnitude was: 1.007337
At 1901 ms, the acceleration's magnitude was: 1.007317
At 1916 ms, the acceleration's magnitude was: 1.010588
At 1934 ms, the acceleration's magnitude was: 1.005163
At 1949 ms, the acceleration's magnitude was: 1.007194
At 1967 ms, the acceleration's magnitude was: 1.008439
At 1982 ms, the acceleration's magnitude was: 1.006901
At 2000 ms, the acceleration's magnitude was: 1.010893
At 2017 ms, the acceleration's magnitude was: 1.002985
At 2036 ms, the acceleration's magnitude was: 1.006933
At 2051 ms, the acceleration's magnitude was: 1.002061
At 2066 ms, the acceleration's magnitude was: 1.004786
At 2088 ms, the acceleration's magnitude was: 1.003911
At 2103 ms, the acceleration's magnitude was: 1.005499
At 2119 ms, the acceleration's magnitude was: 1.004674
At 2134 ms, the acceleration's magnitude was: 1.004902
At 2151 ms, the acceleration's magnitude was: 1.006371
```

Part 3:

```
At 0 minutes, 10 seconds, and 375 milliseconds it was: 1.006650
At 0 minutes, 10 seconds, and 390 milliseconds it was: 1.005347
At 0 minutes, 10 seconds, and 408 milliseconds it was: 1.008235
At 0 minutes, 10 seconds, and 424 milliseconds it was: 1.004897
At 0 minutes, 10 seconds, and 439 milliseconds it was: 1.006890
At 0 minutes, 10 seconds, and 458 milliseconds it was: 1.008470
At 0 minutes, 10 seconds, and 473 milliseconds it was: 1.005878
At 0 minutes, 10 seconds, and 489 milliseconds it was: 1.008925
At 0 minutes, 10 seconds, and 508 milliseconds it was: 1.010528
At 0 minutes, 10 seconds, and 527 milliseconds it was: 1.007864
At 0 minutes, 10 seconds, and 539 milliseconds it was: 1.004561
At 0 minutes, 10 seconds, and 558 milliseconds it was: 1.003831
At 0 minutes, 10 seconds, and 574 milliseconds it was: 1.006732
At 0 minutes, 10 seconds, and 593 milliseconds it was: 1.007075
At 0 minutes, 10 seconds, and 608 milliseconds it was: 1.006416
At 0 minutes, 10 seconds, and 624 milliseconds it was: 1.004486
At 0 minutes, 10 seconds, and 640 milliseconds it was: 1.006197
At 0 minutes, 10 seconds, and 657 milliseconds it was: 1.003409
At 0 minutes, 10 seconds, and 672 milliseconds it was: 1.008906
At 0 minutes, 10 seconds, and 688 milliseconds it was: 1.011073
At 0 minutes, 10 seconds, and 708 milliseconds it was: 1.007197
At 0 minutes, 10 seconds, and 725 milliseconds it was: 1.009637
At 0 minutes, 10 seconds, and 740 milliseconds it was: 1.007433
At 0 minutes, 10 seconds, and 756 milliseconds it was: 1.004761
At 0 minutes, 10 seconds, and 772 milliseconds it was: 1.003706
At 0 minutes, 10 seconds, and 791 milliseconds it was: 1.009728
At 0 minutes, 10 seconds, and 806 milliseconds it was: 1.004745
At 0 minutes, 10 seconds, and 825 milliseconds it was: 1.003087
At 0 minutes, 10 seconds, and 841 milliseconds it was: 1.002389
At 0 minutes, 10 seconds, and 860 milliseconds it was: 1.007844
At 0 minutes, 10 seconds, and 875 milliseconds it was: 1.022746
At 0 minutes, 10 seconds, and 891 milliseconds it was: 1.006313
At 0 minutes, 10 seconds, and 908 milliseconds it was: 1.004397
At 0 minutes, 10 seconds, and 924 milliseconds it was: 1.008718
At 0 minutes, 10 seconds, and 940 milliseconds it was: 1.004029
At 0 minutes, 10 seconds, and 957 milliseconds it was: 1.010965
At 0 minutes, 10 seconds, and 972 milliseconds it was: 1.009976
At 0 minutes, 10 seconds, and 988 milliseconds it was: 1.006040
At 0 minutes, 11 seconds, and 6 milliseconds it was: 1.006974
At 0 minutes, 11 seconds, and 26 milliseconds it was: 1.006842
At 0 minutes, 11 seconds, and 43 milliseconds it was: 1.010233
At 0 minutes, 11 seconds, and 61 milliseconds it was: 1.004536
At 0 minutes, 11 seconds, and 75 milliseconds it was: 1.009190
At 0 minutes, 11 seconds, and 91 milliseconds it was: 1.007958
At 0 minutes, 11 seconds, and 106 milliseconds it was: 1.010189
At 0 minutes, 11 seconds, and 124 milliseconds it was: 1.006526
At 0 minutes, 11 seconds, and 139 milliseconds it was: 1.009974
At 0 minutes, 11 seconds, and 157 milliseconds it was: 1.004986
At 0 minutes, 11 seconds, and 174 milliseconds it was: 1.004678
At 0 minutes, 11 seconds, and 190 milliseconds it was: 1.004994
At 0 minutes, 11 seconds, and 207 milliseconds it was: 1.004929
At 0 minutes, 11 seconds, and 225 milliseconds it was: 1.007091
At 0 minutes, 11 seconds, and 240 milliseconds it was: 1.007128
At 0 minutes, 11 seconds, and 257 milliseconds it was: 1.004349
At 0 minutes, 11 seconds, and 274 milliseconds it was: 1.007408
At 0 minutes, 11 seconds, and 291 milliseconds it was: 1.003320
At 0 minutes, 11 seconds, and 306 milliseconds it was: 1.010045
At 0 minutes, 11 seconds, and 322 milliseconds it was: 1.002067
At 0 minutes, 11 seconds, and 339 milliseconds it was: 1.008839
At 0 minutes, 11 seconds, and 356 milliseconds it was: 1.008380
At 0 minutes, 11 seconds, and 374 milliseconds it was: 1.007712
At 0 minutes, 11 seconds, and 389 milliseconds it was: 1.007163
At 0 minutes, 11 seconds, and 409 milliseconds it was: 1.006361
At 0 minutes, 11 seconds, and 423 milliseconds it was: 1.012118
At 0 minutes, 11 seconds, and 441 milliseconds it was: 1.008008
At 0 minutes, 11 seconds, and 459 milliseconds it was: 1.007379
At 0 minutes, 11 seconds, and 475 milliseconds it was: 1.010892
At 0 minutes, 11 seconds, and 490 milliseconds it was: 1.006869
At 0 minutes, 11 seconds, and 508 milliseconds it was: 1.005328
At 0 minutes, 11 seconds, and 524 milliseconds it was: 1.008490
At 0 minutes, 11 seconds, and 540 milliseconds it was: 1.002519
At 0 minutes, 11 seconds, and 557 milliseconds it was: 1.009428
At 0 minutes, 11 seconds, and 576 milliseconds it was: 1.006990
At 0 minutes, 11 seconds, and 591 milliseconds it was: 1.007732
At 0 minutes, 11 seconds, and 606 milliseconds it was: 1.007519
At 0 minutes, 11 seconds, and 625 milliseconds it was: 1.005343
At 0 minutes, 11 seconds, and 640 milliseconds it was: 1.008820
At 0 minutes, 11 seconds, and 659 milliseconds it was: 1.004451
At 0 minutes, 11 seconds, and 674 milliseconds it was: 1.009128
At 0 minutes, 11 seconds, and 690 milliseconds it was: 1.006235
At 0 minutes, 11 seconds, and 709 milliseconds it was: 1.000220
At 0 minutes, 11 seconds, and 725 milliseconds it was: 1.000906
At 0 minutes, 11 seconds, and 740 milliseconds it was: 1.003998
At 0 minutes, 11 seconds, and 759 milliseconds it was: 1.004975
At 0 minutes, 11 seconds, and 774 milliseconds it was: 1.010649
At 0 minutes, 11 seconds, and 790 milliseconds it was: 1.005932
At 0 minutes, 11 seconds, and 807 milliseconds it was: 1.005137
At 0 minutes, 11 seconds, and 823 milliseconds it was: 1.008748
At 0 minutes, 11 seconds, and 842 milliseconds it was: 1.006804
At 0 minutes, 11 seconds, and 857 milliseconds it was: 1.006123
At 0 minutes, 11 seconds, and 875 milliseconds it was: 1.005083
At 0 minutes, 11 seconds, and 890 milliseconds it was: 1.010302
At 0 minutes, 11 seconds, and 906 milliseconds it was: 1.005569
At 0 minutes, 11 seconds, and 924 milliseconds it was: 1.013671
At 0 minutes, 11 seconds, and 941 milliseconds it was: 1.006381
At 0 minutes, 11 seconds, and 956 milliseconds it was: 1.006103
At 0 minutes, 11 seconds, and 976 milliseconds it was: 1.005241
At 0 minutes, 11 seconds, and 991 milliseconds it was: 1.006186
At 0 minutes, 12 seconds, and 8 milliseconds it was: 1.003920
At 0 minutes, 12 seconds, and 24 milliseconds it was: 1.006242
At 0 minutes, 12 seconds, and 39 milliseconds it was: 1.005279
At 0 minutes, 12 seconds, and 57 milliseconds it was: 1.005846
At 0 minutes, 12 seconds, and 74 milliseconds it was: 1.008306
At 0 minutes, 12 seconds, and 91 milliseconds it was: 1.007747
At 0 minutes, 12 seconds, and 107 milliseconds it was: 1.009119
At 0 minutes, 12 seconds, and 128 milliseconds it was: 1.005467
At 0 minutes, 12 seconds, and 142 milliseconds it was: 1.005525
At 0 minutes, 12 seconds, and 158 milliseconds it was: 1.007151
At 0 minutes, 12 seconds, and 175 milliseconds it was: 1.007055
At 0 minutes, 12 seconds, and 191 milliseconds it was: 1.006282
At 0 minutes, 12 seconds, and 207 milliseconds it was: 1.006504
At 0 minutes, 12 seconds, and 222 milliseconds it was: 1.000570
At 0 minutes, 12 seconds, and 240 milliseconds it was: 1.007759
At 0 minutes, 12 seconds, and 258 milliseconds it was: 1.009702
At 0 minutes, 12 seconds, and 273 milliseconds it was: 1.004419
At 0 minutes, 12 seconds, and 288 milliseconds it was: 1.010871
At 0 minutes, 12 seconds, and 311 milliseconds it was: 1.003768
At 0 minutes, 12 seconds, and 324 milliseconds it was: 1.006745
At 0 minutes, 12 seconds, and 342 milliseconds it was: 1.006189
At 0 minutes, 12 seconds, and 357 milliseconds it was: 1.007194
At 0 minutes, 12 seconds, and 375 milliseconds it was: 1.005906
At 0 minutes, 12 seconds, and 390 milliseconds it was: 1.009529
At 0 minutes, 12 seconds, and 407 milliseconds it was: 1.001619
At 0 minutes, 12 seconds, and 423 milliseconds it was: 1.003117
At 0 minutes, 12 seconds, and 444 milliseconds it was: 1.011808
At 0 minutes, 12 seconds, and 456 milliseconds it was: 1.008267
At 0 minutes, 12 seconds, and 476 milliseconds it was: 1.007800
```

```
simonsj1KCD2048-18 /cygdrive/d/CPRE1850/Lab3
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%
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Part 4:

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